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CS 621 DATA ANALYTICS

Regression Analysis

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PREDICTION(REGRESSION) MODEL

- Within the data science pipeline, the **Analytics Computation** is often is the most important stage.
- Sometimes, the Analytics task is to build a **Prediction Model** out of the data.
- The Model is later used to predict future value.
- If the task is to predict a numerical value, then we need to build a Regression Model.

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REGRESSION/PREDICTION

- Examples:
  - Predict various stock market values.
  - Predict a student's gpa.
  - Predict this year's snow accumulation.
  - Predict your next job's salary.

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LINEAR REGRESSION

- Data with a number of input attributes(features) and an output attribute.

  X1, X2, X3, …. | Y

- Thus, the task is to learn a linear model that predicts a numerical value.

  y = W1*X1 + W2*X2+W3*X3 + …. + W0 + E

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LINEAR REGRESSION

- Linear regression: involves a response variable y and a single predictor variable x

  $y = w_0 + w_1 x$

  where w_0 (y-intercept) and w_1 (slope) are regression coefficients

- Method of least squares: estimates the best-fitting straight line

$$w_1 = \frac{\sum_{i=1}^{|D|}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|}(x_i - \bar{x})^2} \qquad w_0 = \bar{y} - w_1 \bar{x}$$

- Multiple linear regression: involves more than one predictor variable
  - Training data is of the form $(\mathbf{X_1}, y_1)$, $(\mathbf{X_2}, y_2)$,…, $(\mathbf{X_{|D|}}, y_{|D|})$
  - Ex. For 2-D data, we may have: $y = w_0 + w_1 x_1 + w_2 x_2$
  - Solvable by extension of least square method

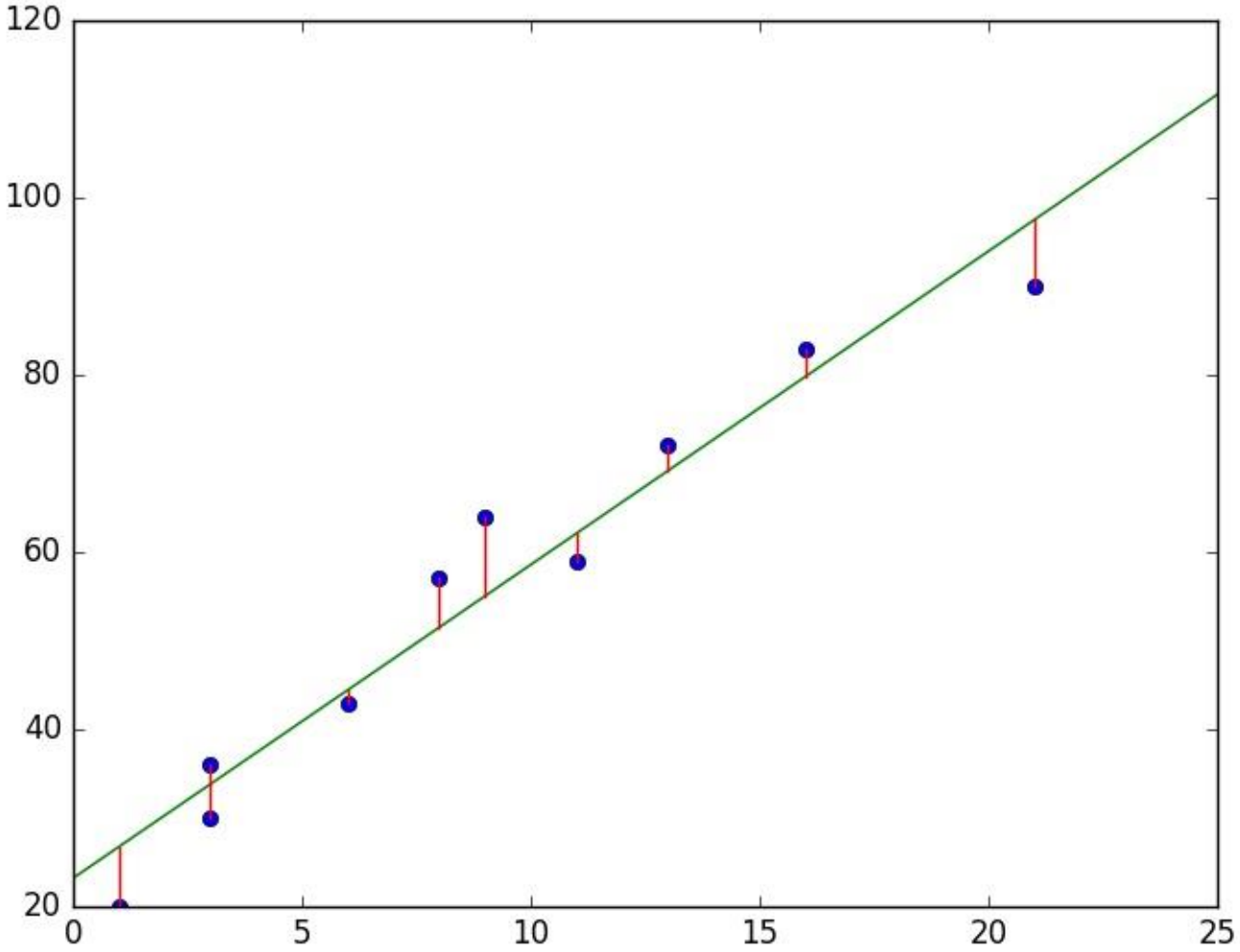
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EXAMPLE

Years of experience	Salary (*1K)
3	30
8	57
9	64
13	72
3	36
6	43
11	59
21	90
1	20
16	83

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REGRESSION LINE



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PREDICTION

X bar = 9.1
Y bar = 55.4
w1 = 3.54
w0 = 23.21

Prediction:
With 10 years of experience, a person will make ~ $58,600.00.

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MODEL ACCURACY/ERROR

- Given training data ; X = [….], Y[….], We constructed a regression model (w_0, w_1) out of X and Y.
- We can apply the regression model on each x of X and have predicted ys, i.e., y'.
- Thus, $\Sigma (y-y')^2$ = sum of errors, is the error/accuracy of the regression model. (the red vertical segments in the previous visualization chart)

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SCIKIT LEARN LINEAR REGRESSION

- Scikit-learn is a Python Machine Learning library.
- Highly popular, especially in academia.
- https://scikit-learn.org/stable/
- We will show how to use its linear regression model in the demo.
- See more at https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

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USE OF LINEAR REGRESSION MODEL

- from sklearn.linear_model import LinearRegression
- import numpy as np
- P = [1.0, 2.0, 3.0, 4.0, 5.0]
- Q = [1.0, 2.0, 1.3, 3.75, 2.25]
- P = np.array(P).reshape((5, 1))
- Q = np.array(Q).reshape((5, 1))
- linreg = LinearRegression()
- linreg.fit(P, Q)
- linreg.predict(R)

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USE OF SKLEARN LINEAR REGRESSION MODEL

- from sklearn.linear_model import LinearRegression
- import numpy as np
- P = [1.0, 2.0, 3.0, 4.0, 5.0]
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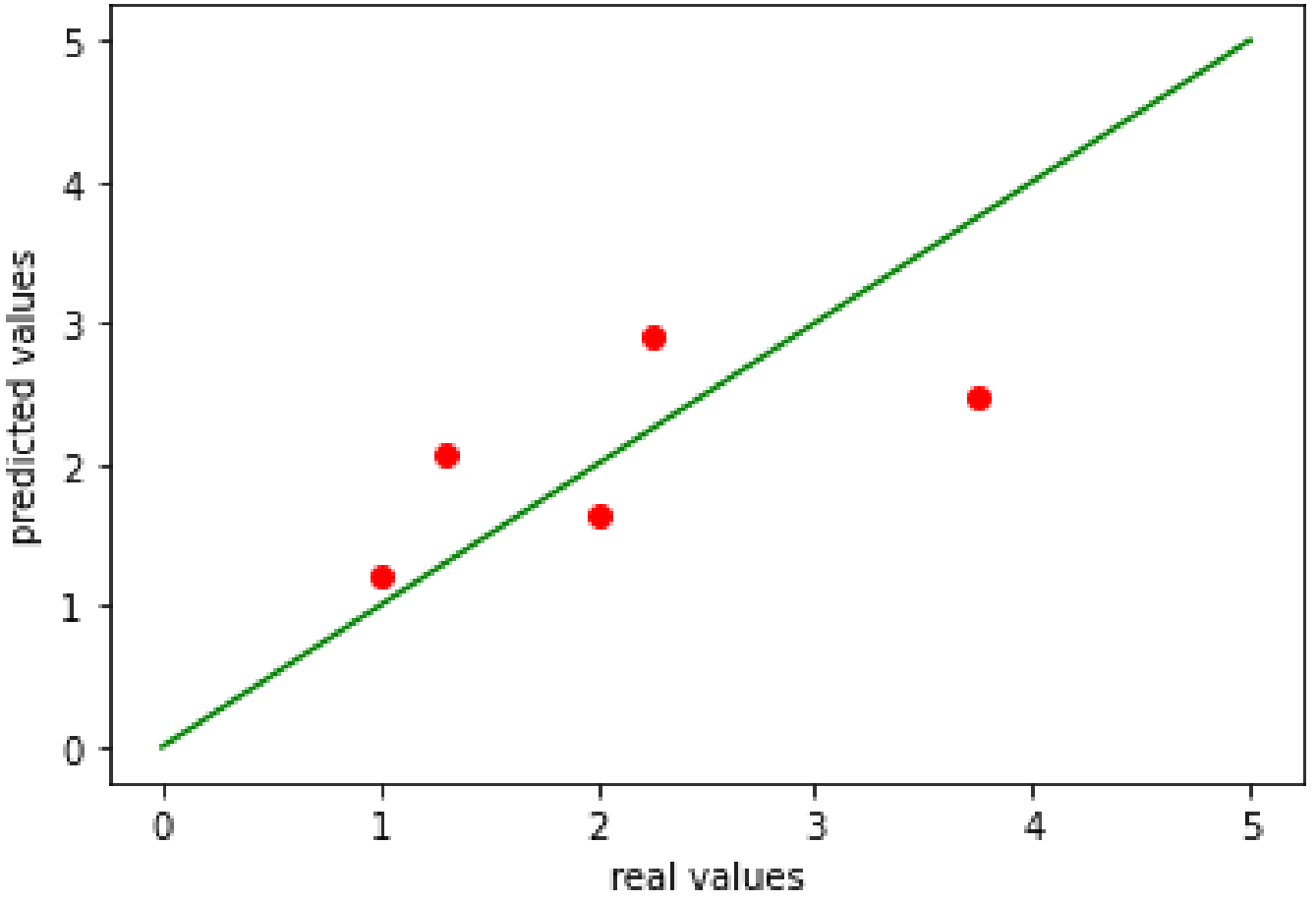
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ERROR ESTIMATION

- from sklearn.metrics import mean_absolute_error, mean_squared_error
- mean_absolute_error(Q, predicted_Q)
- mean_squared_error(Q, predicted_Q)

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TRUE VALUES VS PREDICTIONS





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